

Lab-14.3

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Task 1: AI-Assisted Personal Portfolio Website

Scenario

A student wants to showcase academic projects, technical skills, and contact information through a personal portfolio website. AI tools are used to speed up development while maintaining quality and customization.

Tasks

1. Use GitHub Copilot to generate a semantic HTML structure with sections:

Home

About

Projects

Contact

2. Ask Copilot to suggest responsive CSS using Flexbox or Grid.

3. Customize AI-generated CSS to add:

Hover effects on project cards

Smooth scrolling navigation

Expected Outcome

- Clean and well-structured portfolio website
- Responsive layout for desktop and mobile screens
- Interactive project cards

Code:

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>My Portfolio</title>
7      <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.2/css/all.min.css">
8      <style>
9          body {
10             margin: 0;
11             font-family: 'Arial', sans-serif;
12             background: linear-gradient(to bottom, #1a0933, #3c1b60, #2a1b4d);
13             color: white;
14             overflow-x: hidden;
15         }
16         header {
17             position: relative;
18             height: 100vh;
19             display: flex;
20             flex-direction: column;
21             justify-content: center;
22             align-items: center;
23             text-align: center;
24             background: url('https://via.placeholder.com/1920x1080?text=Space+Background') center/cover no-repeat fixed;
25             padding: 20px;
26         }
27         header h1 {
28             font-size: 3.5em;
29             margin: 0;
30             animation: neon 1.5s infinite alternate;
31         }
32         @keyframes neon {
33             from { text-shadow: 0 0 10px #fff, 0 0 20px #00ddeb, 0 0 30px #00ddeb; }
34             to { text-shadow: 0 0 20px #fff, 0 0 30px #8a2be2, 0 0 40px #8a2be2; }
35         }
36         nav {
37             display: flex;
38             justify-content: center;
39             gap: 15px;
40             margin: 20px 0;
41         }
42         nav a {
```

```

43     color: white;
44     text-decoration: none;
45     font-size: 1.1em;
46     padding: 8px 15px;
47     border: 2px solid #00ddeb;
48     border-radius: 25px;
49     transition: background 0.3s, transform 0.3s;
50 }
51 nav a:hover {
52     background: #00ddeb;
53     color: #1a0933;
54     transform: scale(1.1);
55 }
56 section {
57     padding: 60px 20px;
58     max-width: 1200px;
59     margin: 0 auto;
60 }
61 #about {
62     text-align: center;
63 }
64 .profile-photo {
65     width: 180px;
66     height: 180px;
67     border-radius: 50%;
68     object-fit: cover;
69     border: 4px solid #00ddeb;
70     box-shadow: 0 0 20px #8a2be2, 0 0 40px #00ddeb;
71     margin: 20px auto;
72 }
73 #projects {
74     display: grid;
75     grid-template-columns: repeat(auto-fit, minmax(320px, 1fr));
76     gap: 25px;
77 }
78 .skills {
79     display: grid;
80     grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
81     gap: 20px;
82     background: rgba(255, 255, 255, 0.1);
83     border-radius: 10px;
84     padding: 20px;
85     text-align: center;

```

```

86     transition: transform 0.3s;
87 }
88 .skills:hover {
89     transform: translateY(-10px);
90 }
91 .project-card {
92     perspective: 1000px;
93     height: 300px;
94 }
95 .project-inner {
96     position: relative;
97     width: 100%;
98     height: 100%;
99     transition: transform 0.6s;
100     transform-style: preserve-3d;
101 }
102 .project-card:hover .project-inner {
103     transform: rotate(180deg);
104 }
105 .project-front, .project-back {
106     position: absolute;
107     width: 100%;
108     height: 100%;
109     backface-visibility: hidden;
110     border-radius: 10px;
111     background: rgba(255, 255, 255, 0.1);
112     padding: 20px;
113     text-align: center;
114 }
115 .project-back {
116     transform: rotate(180deg);
117     display: flex;
118     flex-direction: column;
119     justify-content: center;
120 }
121 .project-front img {
122     width: 100%;
123     height: 200px;
124     object-fit: cover;
125     border-radius: 10px;
126 }
127 #contact {
128     text-align: center;

```

```

129     }
130     flex {
131       display: flex;
132       flex-direction: column;
133       max-width: 500px;
134       margin: 0 auto;
135       gap: 12px;
136     }
137     input, textarea {
138       padding: 12px;
139       border: none;
140       border-radius: 6px;
141       background: rgb(255, 255, 255, 0.9);
142     }
143     button {
144       padding: 12px;
145       background: #822922;
146       border: none;
147       color: white;
148       cursor: pointer;
149       border-radius: 6px;
150       border: 1px solid #822922;
151       transition: background 0.3s;
152     }
153     button:hover {
154       background: #000000;
155     }
156     footer {
157       text-align: center;
158       padding: 20px;
159       background: rgb(0, 0, 0, 0.6);
160     }
161     .social-links {
162       margin-top: 15px;
163       display: flex;
164       justify-content: center;
165       gap: 20px;
166     }
167     .social-links a {
168       color: white;
169       font-size: 1.5em;
170       text-decoration: none;
171       padding: 12px;
172       border-radius: 50%;
173       transition: background 0.3s, transform 0.3s;
174     }
175     .social-links a:hover {
176       background: #000000;
177       transform: scale(1.2);
178     }
179     .stars {
180       position: fixed;
181       top: 0;
182       left: 0;
183       width: 100%;
184       height: 100%;
185       pointer-events: none;
186       z-index: -1;

```

```

187     }
188     .star {
189       position: absolute;
190       background: white;
191       border-radius: 50%;
192       animation: twinkle 4s infinite;
193     }
194     @keyframes twinkle {
195       0%, 100% { opacity: 0.3; }
196       50% { opacity: 1; }
197     }
198     </styles>
199     </head>
200     <body>
201       <div class="stars"></div>
202       <header>
203         <h1>Olivia Smith</h1>
204         <nav>
205           <a href="#about">About</a>
206           <a href="#projects">Projects</a>
207           <a href="#contact">Contact</a>
208         </nav>
209       </header>
210       <section id="about">
211         <h2>About Me</h2>
212         
213         <p>Hi, I'm Olivia, a passionate web developer with a flair for creating visually stunning and interactive websites. With 1 year</p>
214       </section>
215       <section id="projects">
216         <h2>My Projects</h2>
217         <div class="project-card">
218           <div class="project-inner">
219             <div class="project-front">
220               <h3>Eye Blinking Sensor</h3>
221             </div>
222             <div class="project-back">
223               <p>An Eye Blinking Sensor using Arduino detects the opening and closing of the eye using an infrared sensor, helps</p>
224             </div>
225           </div>
226         </div>
227         <div class="project-card">
228           <div class="project-inner">
229             <div class="project-front">
230               <h3>Smart Geyser</h3>
231             </div>
232             <div class="project-back">
233               <p>A smart geyser can be controlled remotely using mobile apps or voice assistants. It allows scheduled heating and</p>
234             </div>
235           </div>
236         </div>
237         <div>
238           <h2>Skills</h2>
239           <div class="skills">
240             <div>Cyber Security</div>
241             <div>Networking: TCP/IP, DNS, DHCP, Subnetting, OSI Model, Firewall, VPN</div>
242             <div>Operating Systems: Linux (Kali, Ubuntu), Windows Server, Command Line</div>
243           </div>

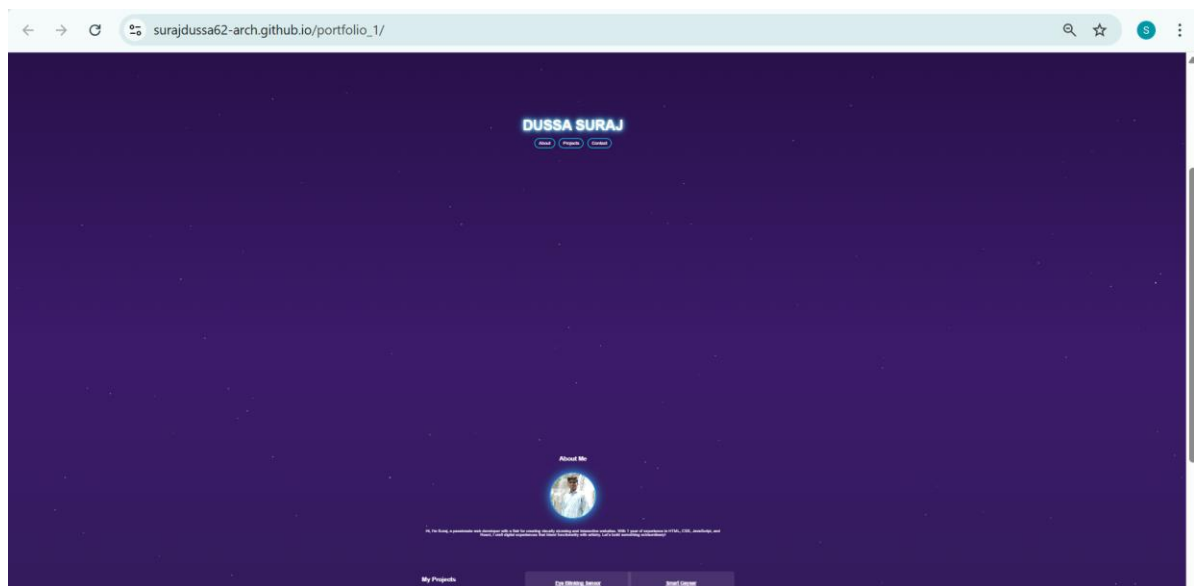
```

```

244
245 <section id="contact">
246   <h2>Connect with Me</h2>
247   <form>
248     <input type="text" placeholder="Your Name">
249     <input type="email" placeholder="Your Email">
250     <textarea placeholder="Your Message"></textarea>
251     <button type="submit">Send Message</button>
252   </form>
253 </section>
254
255 <footer>
256   <p>©2025 Dussa Suraj. All rights reserved. Created with Passion </p>
257   <div class="social-links">
258     <a href="mailto:surajdus@gmail.com" title="Email"><i class="fas fa-envelope"></i></a>
259     <a href="https://www.linkedin.com/in/suraj-dussa-36aa8349/" title="LinkedIn"><i class="fab fa-linkedin"></i></a>
260     <a href="https://github.com/DUSSASURAJ7tab-repositories" title="GitHub"><i class="fab fa-github"></i></a>
261   </div>
262 </footer>
263
264 <script>
265   // Add twinkling stars for cosmic effect
266   const starsContainer = document.querySelector('.stars');
267   for (let i = 0; i < 120; i++) {
268     const star = document.createElement('div');
269     star.classList.add('star');
270     star.style.width = `${Math.random() * 4}px`;
271     star.style.height = star.style.width;
272     star.style.top = `${Math.random() * 100}%`;
273     star.style.left = `${Math.random() * 100}%`;
274     star.style.animationDelay = `${Math.random() * 4}s`;
275     starsContainer.appendChild(star);
276   }
277
278   // Smooth scrolling for nav links
279   document.querySelectorAll('a').forEach(anchor => {
280     anchor.addEventListener('click', function(e) {
281       e.preventDefault();
282       document.querySelector(this.getAttribute('href')).scrollIntoView({
283         behavior: 'smooth'
284       });
285     });
286   });
287 </script>
288 </body>
289
290 </html>
291

```

Output:



Explanation:

HTML Structure

- create a portfolio website with:
 - **Header** (Name + Navigation)
 - **About Section**
 - **Projects Section**
 - **Skills Section**
 - **Contact Form**

- **Footer with Social Links**
-

2 CSS Design

- Used **gradient background** for a cosmic look.
 - Added **neon animation** for your name.
 - Created **hover effects** for buttons.
 - Used **flip card animation** for projects.
 - Added **glowing circular profile photo**.
 - Designed responsive grid layout.
-

3 JavaScript Features

- Created **twinkling stars background**.
- Added **smooth scrolling** when clicking navigation links.

Output:

When you open the file in a browser, you see:

- Full-screen header with glowing **DUSSA SURAJ**
- Navigation buttons (About, Projects, Contact)
- About section with circular photo
- Project cards that flip on hover
- Skills section
- Contact form
- Animated twinkling stars background
- Social media icons in footer

Task 2: AI-Generated Restaurant Landing Page

Scenario

A local restaurant requires a simple yet attractive landing page to display its offerings. The developer uses AI assistance to rapidly generate UI components.

Tasks

1. Use Copilot to create a navigation bar with links:

Home

Menu

Gallery

Contact

2. Generate a menu section styled with CSS cards.

3. Implement a JavaScript-based image slider for the gallery using Copilot suggestions.

4. Customize animations or transitions for better user experience.

Expected Outcome

- Fully functional restaurant landing page
- Responsive navigation bar
- Working image slider with smooth transitions

Code:

```

1 |<!DOCTYPE html>
2 |<html lang="en">
3 |<head>
4 |<meta charset="UTF-8">
5 |<meta name="viewport" content="width=device-width, initial-scale=1.0">
6 |<title>Elegant Restaurant</title>
7 |
8 |<style>
9 |{
10 |    margin:0;
11 |    padding:0;
12 |    box-sizing:border-box;
13 |    font-family: 'Segoe UI', sans-serif;
14 |}
15 |
16 |html{
17 |    scroll-behavior:smooth;
18 |}
19 |
20 |body{
21 |    background: #f1f1f1;
22 |    color: #333;
23 |}
24 |
25 |/* NAVBAR */
26 |.navbar{
27 |    background: #333;
28 |    padding: 20px;
29 |    position: fixed;
30 |    width: 100%;
31 |    top: 0;
32 |    z-index: 1000;
33 |}
34 |
35 |.nav-menu{
36 |    display: flex;
37 |    justify-content: center;
38 |    gap: 40px;
39 |    list-style: none;
40 |}
41 |
42 |.nav-menu a{
43 |    color: #fff;
44 |    text-decoration: none;
45 |    font-size: 18px;
46 |    transition: 0.3s;
47 |}
48 |
49 |.nav-menu a:hover{
50 |    color: #d4af37;
51 |}
52 |
53 |/* HERO */
54 |#home{
55 |    height: 100vh;
56 |    background: url("https://images.unsplash.com/photo-1517248135467-4c7edcad34c4") center/cover no-repeat;
57 |    display: flex;
58 |    flex-direction: column;
59 |    justify-content: center;
60 |    align-items: center;
61 |    text-align: center;
62 |    padding-top: 80px;
63 |}
64 |
65 |#home h1{

```

```

2 <html lang="en">
3 <head>
8 <style>
54 #home{
61 |   text-align:center;
62 |   padding-top:80px;
63 | }
64 #home h1{
65 |   font-size:50px;
66 |   margin-bottom:20px;
67 | }
68
69 #home p{
70 |   font-size:22px;
71 |   max-width:600px;
72 | }
73
74 /* SECTIONS */
75 section{
76 |   padding:100px 20px;
77 |   text-align:center;
78 | }
79
80 h2{
81 |   font-size:35px;
82 |   margin-bottom:30px;
83 |   color:#0d4af37;
84 | }
85
86 /* MENU */
87 .menu-items div{
88 |   margin:12px 0;
89 |   font-size:20px;
90 | }
91
92 /* GALLERY */
93 .gallery-grid{
94 |   display:grid;
95 |   grid-template-columns:repeat(auto-fit,minmax(250px,1fr));
96 |   gap:20px;
97 |   margin-top:30px;
98 | }
99
100 .gallery-grid img{
101 |   width:100%;
102 |   height:250px;
103 |   object-fit:cover;
104 |   border-radius:10px;
105 |   transition:0.4s;
106 | }
107
108 .gallery-grid img:hover{
109 |   transform:scale(1.05);
110 | }
111
112 /* CONTACT */
113 .contact-box{
114 |   max-width:600px;
115 |   margin:auto;
116 |   background:#f1c1c1;
117 |   padding:30px;
118 |   border-radius:10px;
119 | }
120
121 .contact-box p{

```



```

122     margin:10px 0;
123     font-size:18px;
124 }
125
126 /* FOOTER */
127 footer{
128     background: #f1f1f1;
129     text-align:center;
130     padding:25px;
131     margin-top:50px;
132 }
133 </style>
134 </head>
135
136 <body>
137
138 <nav class="navbar">
139     <ul class="nav-menu">
140         <li><a href="#home">Home</a></li>
141         <li><a href="#menu">Menu</a></li>
142         <li><a href="#gallery">Gallery</a></li>
143         <li><a href="#contact">Contact</a></li>
144     </ul>
145 </nav>
146
147 <!-- HOME -->
148 <section id="home">
149     <h1>Welcome to Elegant Restaurant</h1>
150     <p>Experience luxury dining with exquisite flavors and exceptional service.</p>
151 </section>
152
153 <!-- MENU -->
154 <section id="menu">
155     <h2>Our Special Menu</h2>
156     <div class="menu-items">
157         <div>🥩 Grilled Steak - $25</div>
158         <div>🍝 Creamy Alfredo Pasta - $18</div>
159         <div>🍕 Wood Fired Pizza - $16</div>
160         <div>🍣 Sushi Platter - $22</div>
161         <div>🍰 Chocolate Lava Cake - $9</div>
162     </div>
163 </section>
164
165 <!-- GALLERY -->
166 <section id="gallery">
167     <h2>Our Gallery</h2>
168     <div class="gallery-grid">
169         
170         
171         
172         
173     </div>
174 </section>
175
176 <!-- CONTACT -->
177 <section id="contact">
178     <h2>Contact Us</h2>
179     <div class="contact-box">
180         <p>📍 123 Fine Dining Street, Culinary City</p>
181         <p>📞 +1 234 567 890</p>
182         <p>✉ info@elegantrestaurant.com</p>
183     </div>

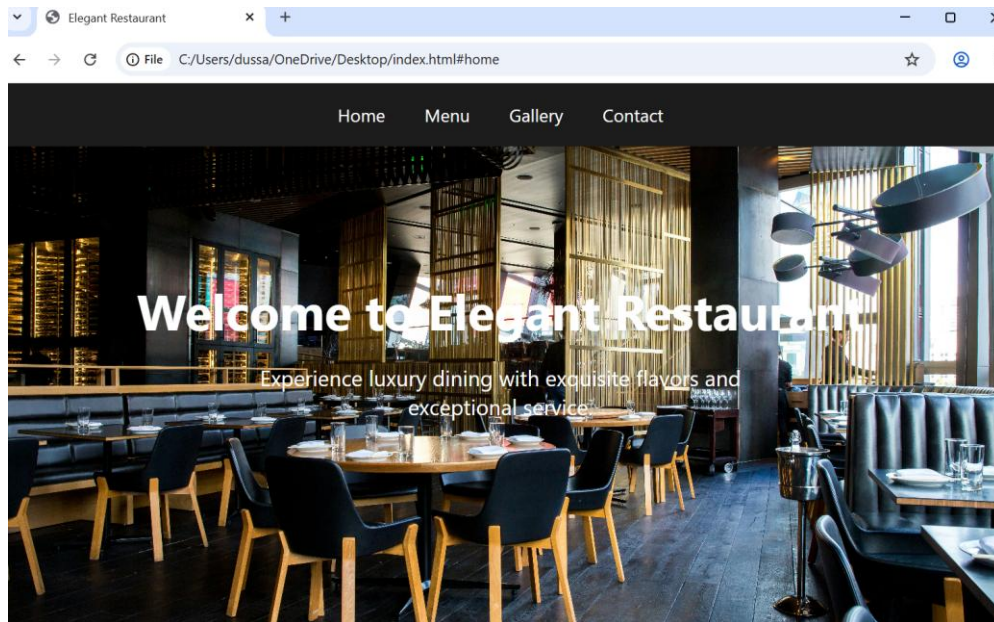
```

```

175 <!-- CONTACT -->
176 <section id="contact">
177     <h2>Contact Us</h2>
178     <div class="contact-box">
179         <p>📍 123 Fine Dining Street, Culinary City</p>
180         <p>📞 +1 234 567 890</p>
181         <p>✉ info@elegantrestaurant.com</p>
182         <p>🕒 Mon - Sun: 10 AM - 11 PM</p>
183     </div>
184 </section>
185
186
187 <footer>
188     <p>© 2026 Elegant Restaurant. All Rights Reserved.</p>
189 </footer>
190
191 </body>
192 </html>
193

```

Output:



Explanation:

- Navbar → Fixed dark menu bar with smooth scrolling links (Home, Menu, Gallery, Contact).
- Theme → Dark luxury design with gold accent color for a premium look.
- Hero Section → Full-screen background image with welcome text.
- Menu Section → Displays special food items with prices.
- Gallery Section → Food and restaurant images in grid layout with hover zoom effect.
- Contact Section → Shows address, phone number, email, and opening hours inside a styled box.
- Footer → Simple copyright section at bottom.

Task 3: AI-Powered Event Registration Form

Scenario

SR University is hosting a technical fest and requires an online registration form with real-time validation to ensure accurate user input.

Tasks

1. Ask Copilot to generate an HTML form with the following fields:

Name

Email

Phone Number

Department

Event Selection

2. Use AI assistance to style the form using CSS for readability and accessibility.

3. Implement JavaScript validation using Copilot:

Email format validation

Phone number length validation

Required field checks

Expected Outcome

- User-friendly and visually appealing form
- Real-time validation messages
- Prevention of invalid submissions

Code:

```
1 <!doctype html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Event Registration Form</title>
7   <style>
8     body {
9       font-family: Arial, sans-serif;
10      max-width: 500px;
11      margin: 50px auto;
12      padding: 20px;
13      background-color: #f9f9f9;
14    }
15
16    .form-container {
17      background: #fff;
18      padding: 30px;
19      border-radius: 8px;
20      box-shadow: 0 2px 10px 0 rgba(0,0,0,0.1);
21    }
22
23    h1 {
24      color: #e333;
25      text-align: center;
26    }
27
28    .form-group {
29      margin-bottom: 20px;
30    }
31
32    label {
33      display: block;
34      margin-bottom: 5px;
35      font-weight: bold;
36      color: #e333;
37    }
38
39    input, select {
40      width: 100%;
41      padding: 10px;
42      border: 2px solid #ddd;
43      border-radius: 4px;
44      font-size: 14px;
45      box-sizing: border-box;
46    }
47
48    input:focus, select:focus {
49      outline: none;
50      border-color: #e333;
51    }
52
53    .error {
54      border-color: #d32f2f !important;
55    }
56
57    .error-message {
58      color: #d32f2f;
59      font-size: 12px;
60      margin-top: 5px;
61      display: none;
62    }
63
64    .success-message {
65      color: #e333;
66    }
```

```

66     font-size: 12px;
67     margin-top: 5px;
68     display: none;
69 }
70
71 button {
72     width: 100%;
73     padding: 10px;
74     background-color: #007bff;
75     color: white;
76     border: none;
77     border-radius: 5px;
78     font-size: 16px;
79     font-weight: bold;
80     cursor: pointer;
81     transition: background-color 0.3s;
82 }
83
84 button:hover {
85     background-color: #0056b3;
86 }
87
88 button:disabled {
89     background-color: #cccccc;
90     cursor: not-allowed;
91 }
92
93 </style>
94 </head>
95 <body>
96     <div class="form-container">
97         <h1>Event Registration</h1>
98         <form id="registrationForm">
99             <div class="form-group">
100                 <label for="name">Name </label>
101                 <input type="text" id="name" name="name" required>
102                 <div class="error-message" id="nameError">Name must be at least 2 characters</div>
103                 <div class="success-message" id="nameSuccess"></div>
104             </div>
105
106             <div class="form-group">
107                 <label for="email">Email </label>
108                 <input type="email" id="email" name="email" required>
109                 <div class="error-message" id="emailError">Enter a valid email address</div>
110                 <div class="success-message" id="emailSuccess"></div>
111             </div>
112
113             <div class="form-group">
114                 <label for="phone">Phone </label>
115                 <input type="tel" id="phone" name="phone" required>
116                 <div class="error-message" id="phoneError">Enter a valid phone number (10 digits)</div>
117                 <div class="success-message" id="phoneSuccess"></div>
118             </div>
119
120             <div class="form-group">
121                 <label for="department">Department </label>
122                 <select id="department" name="department" required>
123                     <option value="">Select Department</option>
124                     <option value="hr">Human Resources</option>
125                     <option value="it">IT</option>
126                     <option value="sales">Sales</option>

```

```

127                     <option value="marketing">Marketing</option>
128                     <option value="other">Other</option>
129                 </select>
130                 <div class="error-message" id="departmentError">Please select a department</div>
131                 <div class="success-message" id="departmentSuccess"></div>
132             </div>
133
134             <div class="form-group">
135                 <label for="event">Event Selection </label>
136                 <select id="event" name="event" required>
137                     <option value="">Select Event</option>
138                     <option value="conference">Annual Conference</option>
139                     <option value="workshop">Technical Workshop</option>
140                     <option value="webinar">Webinar</option>
141                     <option value="networking">Networking Dinner</option>
142                 </select>
143                 <div class="error-message" id="eventError">Please select an event</div>
144                 <div class="success-message" id="eventSuccess"></div>
145             </div>
146
147             <button type="submit" id="submitBtn">Register</button>
148         </form>
149     </div>
150
151     <script>
152         const form = document.getElementById('registrationForm');
153         const nameInput = document.getElementById('name');
154         const emailInput = document.getElementById('email');
155         const phoneInput = document.getElementById('phone');
156         const departmentSelect = document.getElementById('department');
157         const eventSelect = document.getElementById('event');
158         const submitBtn = document.getElementById('submitBtn');
159
160         const validators = {
161             name: (value) => value.trim().length >= 2,
162             email: (value) => /^[a-zA-Z0-9]+@[a-zA-Z0-9]+\.[a-zA-Z]{2,}$/.test(value),
163             phone: (value) => /^[0-9]{10}$/.test(value.replace(/\D/g, '')),
164             department: (value) => value !== '',
165             event: (value) => value !== ''
166         };
167
168         function validateField(fieldName, value) {
169             const isValid = validators[fieldName](value);
170             const fieldElement = document.getElementById(fieldName);
171             const errorElement = document.getElementById(fieldName + 'Error');
172             const successElement = document.getElementById(fieldName + 'Success');
173
174             if (isValid) {
175                 fieldElement.classList.remove('error');
176                 errorElement.style.display = 'none';
177                 successElement.style.display = 'block';
178             } else {
179                 fieldElement.classList.add('error');
180                 errorElement.style.display = 'block';
181                 successElement.style.display = 'none';
182             }
183
184             return isValid;
185         }
186     </script>

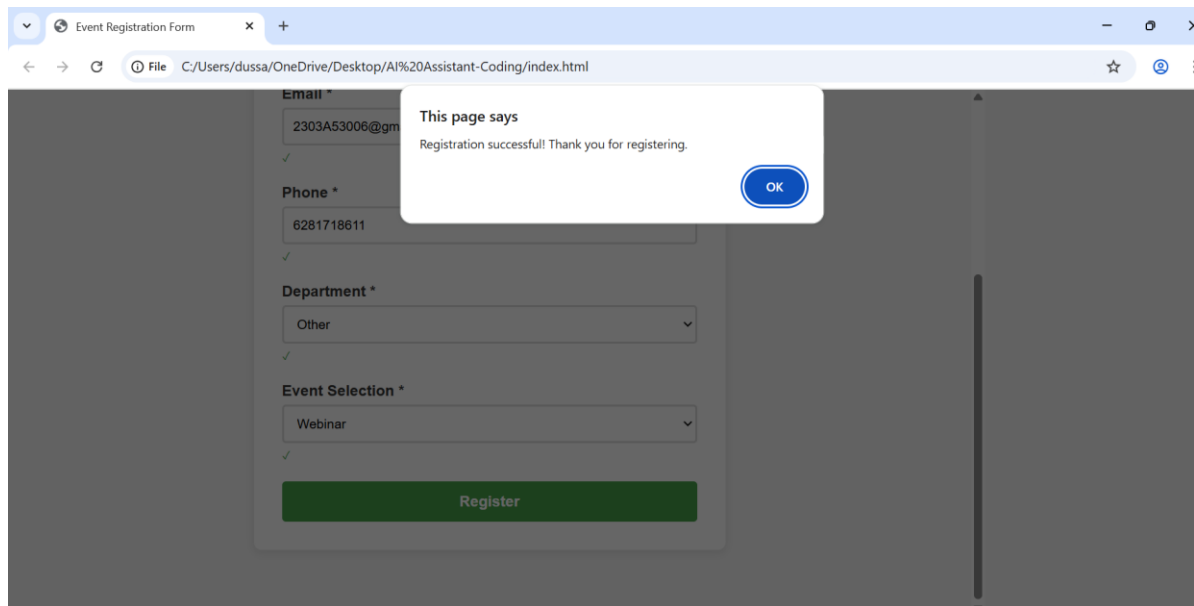
```

```

187
188     function checkFormValidity() {
189         const allValid =
190             validateField('name', nameInput.value) &&
191             validateField('email', emailInput.value) &&
192             validateField('phone', phoneInput.value) &&
193             validateField('department', departmentSelect.value) &&
194             validateField('event', eventSelect.value);
195
196         submitBtn.disabled = !allValid;
197     }
198
199     nameInput.addEventListener('input', (e) => validateField('name', e.target.value) && checkFormValidity());
200     emailInput.addEventListener('input', (e) => validateField('email', e.target.value) && checkFormValidity());
201     phoneInput.addEventListener('input', (e) => validateField('phone', e.target.value) && checkFormValidity());
202     departmentSelect.addEventListener('change', (e) => validateField('department', e.target.value) && checkFormValidity());
203     eventSelect.addEventListener('change', (e) => validateField('event', e.target.value) && checkFormValidity());
204
205     form.addEventListener('submit', (e) => {
206         e.preventDefault();
207         alert('Registration successful! Thank you for registering.');
208         form.reset();
209         submitBtn.disabled = true;
210         document.querySelector('.success-message').forEach(el => el.style.display = 'none');
211     });
212
213     checkFormValidity();
214 </script>
215 </body>
216 </html>

```

Output:



Explanation:

- HTML Structure → Creates an Event Registration Form with Name, Email, Phone, Department, and Event fields.
- CSS Styling → Adds clean white card design, centered layout, rounded corners, shadow, and green theme.
- Required Fields → All inputs are mandatory (required).
- Error Messages → Hidden by default; shown if input is invalid.
- Success Message (✓) → Displays when a field is valid.
- JavaScript Validators →
 - Name → Minimum 2 characters
 - Email → Valid email format using regex
 - Phone → Exactly 10 digits
 - Department & Event → Must be selected
- Real-time Validation → Checks inputs while typing (input & change events).
- Submit Button → Disabled until all fields are valid.
- Form Submit → Shows success alert, resets form, disables button again.

Task 4: AI-Assisted E-Commerce Product Page

Scenario

A startup is developing a basic e-commerce product page to showcase products and manage simple cart functionality.

Tasks

1. Use Copilot to generate a grid-based product catalog using HTML and CSS.
2. Implement an “Add to Cart” feature using JavaScript.
3. Modify AI-generated code to include:
Cart item counter at the top-right corner
Visual feedback when a product is added

Expected Outcome

- Functional product catalog
- Working cart counter
- Interactive user experience

Code:

```

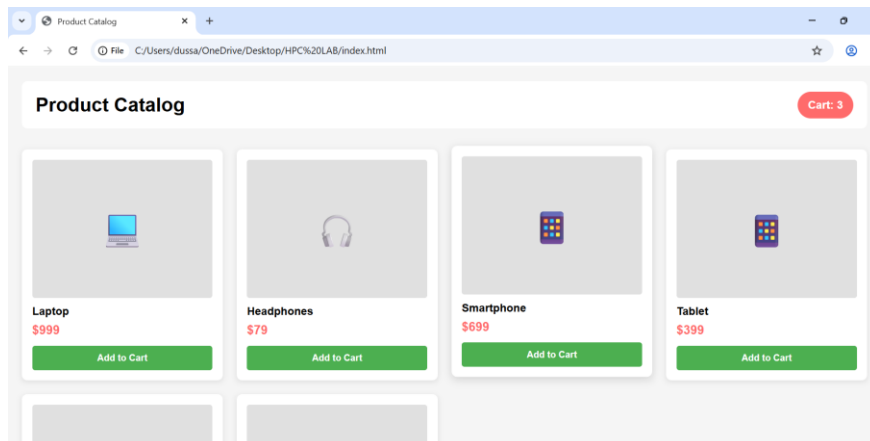
<!-- CSS Start -->
<!-- Tailwind -->
<!-- Base -->
<!-- meta charset="utf-8" -->
<!-- meta name="viewport" content="width=device-width, initial-scale=1.0" -->
<!-- title=Product Catalog -->
<!-- style -->
<!-- { -->
<!--   margin: 0; -->
<!--   padding: 0; -->
<!--   box-sizing: border-box; -->
<!-- } -->
<!-- body -->
<!-- font-family: Arial, sans-serif; -->
<!-- background-color: #f0f0f0; -->
<!-- padding: 20px; -->
<!-- } -->
<!-- header -->
<!-- display: flex; -->
<!-- justify-content: space-between; -->
<!-- align-items: center; -->
<!-- margin-bottom: 10px; -->
<!-- background-color: #fff; -->
<!-- padding: 5px 10px; -->
<!-- border-radius: 5px; -->
<!-- } -->
<!-- h1 -->
<!-- font-size: 24px; -->
<!-- } -->
<!-- .cart-counter -->
<!-- background-color: #f0f0f0; -->
<!-- color: #fff; -->
<!-- padding: 5px 10px; -->
<!-- border-radius: 5px; -->
<!-- font-weight: bold; -->
<!-- font-size: 16px; -->
<!-- } -->
<!-- .catalog -->
<!-- display: grid; -->
<!-- grid-template-columns: repeat(auto-fill, minmax(250px, 1fr)); -->
<!-- gap: 20px; -->
<!-- } -->
<!-- .product-card -->
<!-- background-color: #fff; -->
<!-- border-radius: 10px; -->
<!-- padding: 10px; -->
<!-- box-shadow: 0 2px 4px #ccc; -->
<!-- transition: transform 0.2s; -->
<!-- } -->
<!-- .product-card:hover -->
<!-- transform: translate(0, -5px); -->
<!-- box-shadow: 0 4px 8px #ccc; -->
<!-- } -->
<!-- .product-image -->
<!-- width: 100%; -->
<!-- height: 200px; -->
<!-- background-color: #ccc; -->
<!-- border-radius: 5px; -->
<!-- margin-bottom: 10px; -->
<!-- display: flex; -->
<!-- align-items: center; -->
<!-- justify-content: center; -->
<!-- font-size: 48px; -->
<!-- } -->
<!-- .product-name -->
<!-- font-size: 16px; -->
<!-- font-weight: bold; -->
<!-- margin-bottom: 5px; -->
<!-- } -->
```

```

40 .product-price {
41   font-size: 18px;
42   color: #ff6600;
43   font-weight: bold;
44   margin-bottom: 10px;
45 }
46
47 .btn-add {
48   width: 100%;
49   padding: 10px;
50   background-color: #f0f0f0;
51   color: #000;
52   border: none;
53   border-radius: 4px;
54   cursor: pointer;
55   font-weight: bold;
56   transition: background-color 0.3s;
57 }
58
59 .btn-add:hover {
60   background-color: #e0e0e0;
61 }
62
63 .btn-add:active {
64   background-color: #d0d0d0;
65 }
66
67 //style
68 </head>
69 <body>
70   <header>
71     <h1>Product Catalog</h1>
72     <div class="cart-counter">Cart: <span id="cartCount">0</span></div>
73   </header>
74
75   <div class="catalog" id="catalog"></div>
76
77   <script>
78     const products = [
79       { id: 1, name: "Laptop", price: "$999", emoji: "💻" },
80       { id: 2, name: "Headphones", price: "$79", emoji: "🎧" },
81       { id: 3, name: "Smartphone", price: "$699", emoji: "📱" },
82       { id: 4, name: "Tablet", price: "$399", emoji: "📺" },
83       { id: 5, name: "Camera", price: "$599", emoji: "📷" },
84       { id: 6, name: "Watch", price: "$199", emoji: "🕒" },
85     ];
86
87     let cartCount = 0;
88
89     function renderCatalog() {
90       const catalog = document.getElementById("catalog");
91       catalog.innerHTML = products.map(product => `
92         <div class="product-card">
93           <div class="product-image"><img alt="${product.emoji}" /></div>
94           <div class="product-name"><h3>${product.name}</h3></div>
95           <div class="product-price"><h2>${product.price}</h2></div>
96           <button class="btn-add" onclick="addToCart(${product.id})">Add to Cart</button>
97         </div>
98       `).join("");
99
100       function addToCart(productId) {
101         cartCount++;
102         document.getElementById("cartCount").textContent = cartCount;
103       }
104
105       renderCatalog();
106     }
107   </script>
108 </body>
109 </html>

```

Output:



Explanation:

- HTML Structure → Creates a Product Catalog page with header and product grid.
- Header Section → Displays page title and cart counter (Cart: 0).
- CSS Styling →
 - Grid layout for products
 - White product cards with shadow
 - Hover effect (card lifts up)
 - Green “Add to Cart” button

- Red cart counter badge
- Products Array (JavaScript) → Stores product details (id, name, price, emoji).
- renderCatalog() Function → Dynamically displays all products using map() and innerHTML.
- addToCart() Function → Increases cart count and updates it on the page.
- onclick Event → Adds selected product to cart when button is clicked.
- Initial Call → renderCatalog() loads products when page opens.

Task 5: Create a Responsive Web Page Layout

Instructions:

- Design a basic web page layout with a header, main content area, and footer using HTML and CSS.
- Use AI to assist in generating responsive CSS for different screen sizes.
- Ensure the layout is clean and visually organized.

Expected Output:

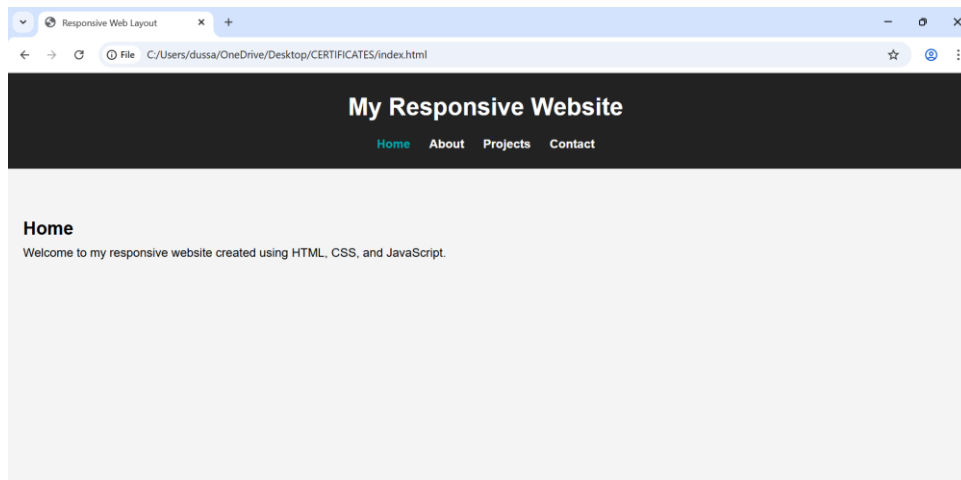
- A responsive web page with:
- Header with navigation links
- Main section with placeholder text/images
- Footer with copyright or contact info
- Layout adapts correctly to desktop, tablet, and mobile screen sizes.

Code:


```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <title>Responsive Web Layout</title>
6   <meta name="viewport" content="width=device-width, initial-scale=1.0">
7
8   <style>
9     * {
10       margin: 0;
11       padding: 0;
12       box-sizing: border-box;
13       font-family: Arial, sans-serif;
14     }
15
16     body {
17       line-height: 1.6;
18     }
19
20     header {
21       background: #222;
22       color: white;
23       padding: 20px 0;
24       text-align: center;
25     }
26
27     nav {
28       display: flex;
29       justify-content: center;
30       gap: 25px;
31       margin-top: 10px;
32     }
33
34     nav a {
35       color: white;
36       text-decoration: none;
37       font-weight: bold;
38     }
39
40     nav a.active {
41       color: #00adb5;
42     }
43
44     section {
45       padding: 60px 20px;
46       min-height: 100vh;
47     }
48
49     #home { background: #f4f4f4; }
50     #about { background: #e3f2fd; }
51     #projects { background: #fce4ec; }
52     #contact { background: #f1f8e9; }
53
54     .project-card, .skill-tag {
55       background: white;
56       padding: 20px;
57       margin: 10px 0;
58       border-radius: 8px;
59       box-shadow: 0 4px 8px rgba(0,0,0,0.1);
60     }
61
62     footer {
63       background: #222;
64       color: white;
65       text-align: center;
66       padding: 15px;
67     }
68
69     /* Responsive Layout */
70     @media (max-width: 768px) {
71       nav {
72         flex-direction: column;
73         align-items: center;
74       }
75
76       section {
77         padding: 40px 15px;
78       }
79     }
80   </style>
81 </head>
82
83 <body>
84
85   <header id="header">
86     <h1>My Responsive Website</h1>
87     <nav>
88       <a href="#home" class="nav-link">Home</a>
89       <a href="#about" class="nav-link">About</a>
90       <a href="#projects" class="nav-link">Projects</a>
91       <a href="#contact" class="nav-link">Contact</a>
92     </nav>
93   </header>
94
95   <section id="home">
96     <h2>Home</h2>

```

Explanation

- The HTML creates a page with header, navigation, sections, contact form, and footer.
- CSS uses Flexbox for navigation and a media query to make the layout responsive.
- JavaScript enables smooth scrolling when clicking menu links.
- The contact form includes basic validation for empty fields and email format.
- Active navigation links highlight based on scroll position.
- Scroll animations make project cards and skills appear smoothly.